

Working with Executors and Thread Pools



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java.util.concurrent Library

**Facilitates multithreading
programming**

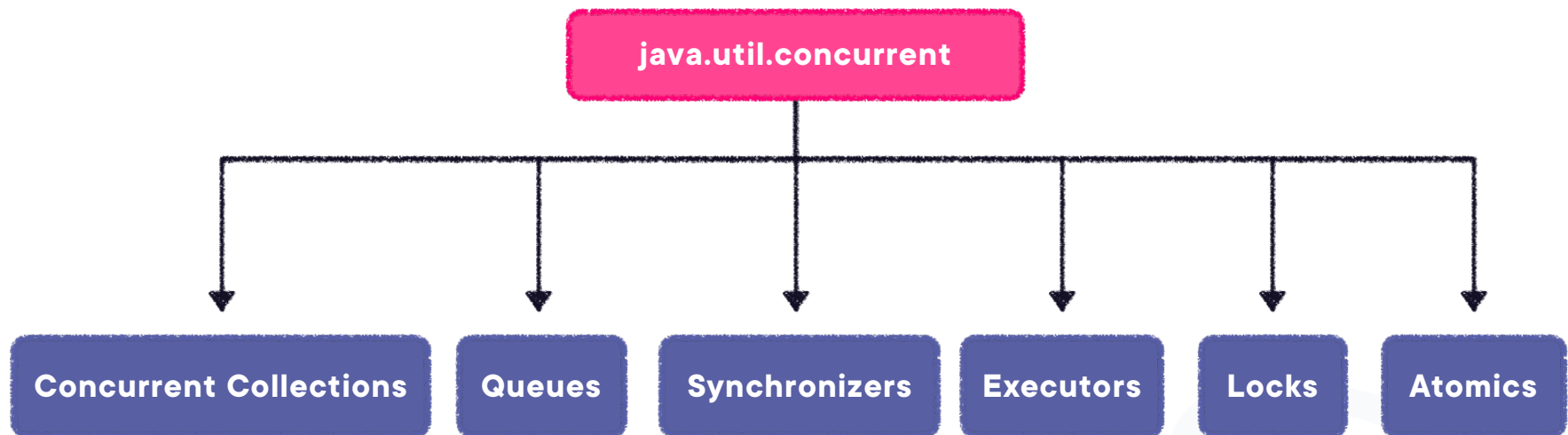
Introduced in Java 5

Multiple packages

**Simplifies the implementation of
concurrency**

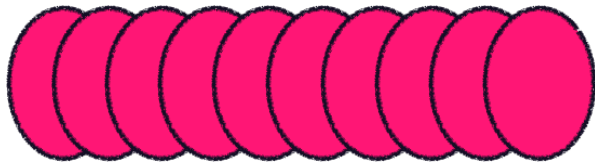


java.util.concurrent Library

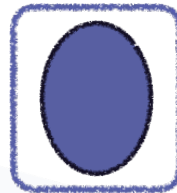
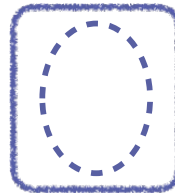
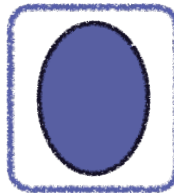
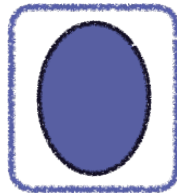
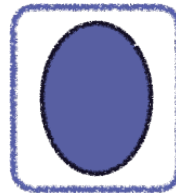


Executors: Thread Pools

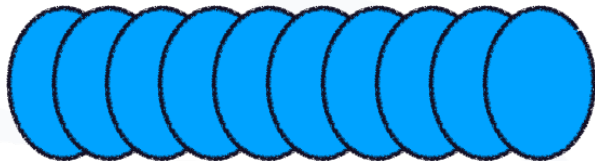
Task Queue



Thread Pool



Completed Tasks

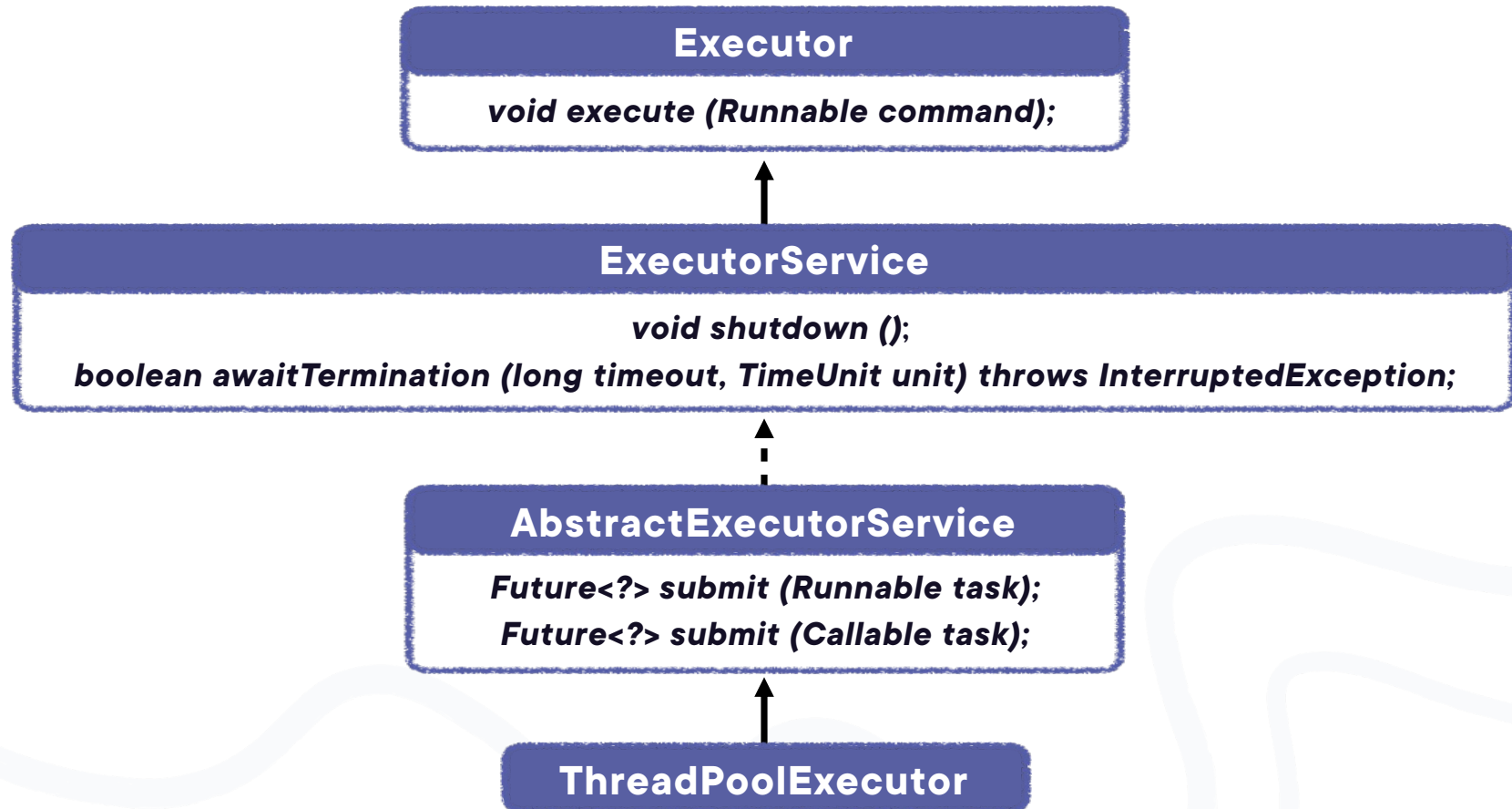


The Runnable Interface

```
@FunctionalInterface
public interface Runnable {
    /**
     * Runs this operation.
     */
    void run();
}
```



Using Executors





Demo: Executors Demo





Using the Callable Interface



The Callable Interface

```
@FunctionalInterface
public interface Callable<V> {
    /**
     * Computes a result, or throws an
     * exception if unable to do so.
     *
     * @return computed result
     * @throws Exception if unable to
     * compute a result
     */
    V call() throws Exception;
}
```



Working with the Callable Interface

Callable<V>

**submit(Callable<V>)
-> Future<V>**

future.get()

future.isDone()

future.cancel()





Demo: Callable Demo

